Computer Networks Lab 11

Course: Computer Networks (CL3001) | Semester: Fall 2026
Course Coordinator: Ms. Ramsha Iqbal

Learning Objectives

- Introduction to Wireless Network.
- Configuration of Wireless Module (WRT300N) on PCs.
- Introduction and configuration of NAT & its Types.
- Wireshark of NAT.

Wireless Network

1. Introduction to Wireless Network

A wireless network enables devices to maintain network connectivity while roaming, without requiring physical cabling. Access points extend the coverage of Wi-Fi signals, allowing a computer to remain connected even at a significant distance from the router. When connecting to a Wi-Fi hotspot in a café or other public location, you are accessing that organization's wireless network.

The fundamental distinction between wireless and wired networks lies in their method of connectivity. A wired network requires a physical connection typically an Ethernet cable linking a device, such as a laptop or desktop computer, to a network router to access the Internet or another network. This physical dependency limits mobility, as the device must remain tethered to the router, which is considered a primary disadvantage compared to the flexibility of a wireless network.

2. Importance of Wireless Technology

However, delving into a specific technology at this point is getting ahead of the tale. Regardless of how the protocols are constructed or what sort of data they carry; wireless networks have some important advantages. The most obvious benefit of wireless networking is mobility. Users of wireless networks can connect to existing networks and then roam freely. Because the phone connects the user via cell towers, a mobile phone user can travel kilometers in a single call.

Initially, mobile telephone was prohibitively expensive. Due to the exorbitant price, it was only used by highly mobile professionals such as sales managers and important executive decision-makers who needed to be accessible at any time and from any location. Mobile telephone, on the other hand, has proven to be a valuable service that is becoming increasingly popular. Wireless networks often provide a lot of flexibility, which translates to quick installation. Wireless networks link users to an existing network using a variety of base stations.

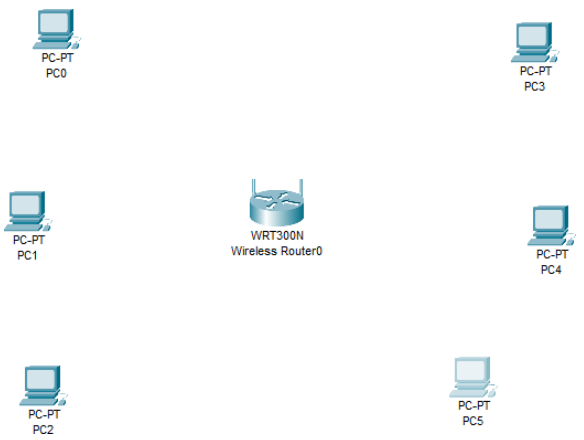
3. Working of Wireless Network

A wireless local area network (WLAN) connects a collection of computers in the same way that a wired network does. Because “wireless” does not require costly cabling, the major benefit is that it is generally easier, faster, and less expensive to set up. Building a network by dragging cables over an office's walls and ceilings, on the other hand, may be time- consuming and costly. Even if you currently have a wired network, a wireless network might be a cost- effective method to extend or expand it.

Radio Frequency (RF) technology, a frequency related with radio wave transmission within the electromagnetic spectrum, is used to power wireless networks. When an RF current is sent into an antenna, it creates an electromagnetic field that can travel over space. A wireless network's core is a mechanism known as an access point (AP). The fundamental function of an access point is to transmit a wireless signal that computers can detect and tune into. Because wireless networks are frequently linked to wired networks, access points frequently serve as a portal to the resources of the wired network, such as an Internet connection.

To connect to an access point and join a wireless network computers must be equipped with wireless network adapters. These are usually incorporated into the device but if not, any computer or notebook can be turned wireless-capable by connecting an add-on adapter to an empty expansion slot, USB port or in case of notebooks, a PC card slot.

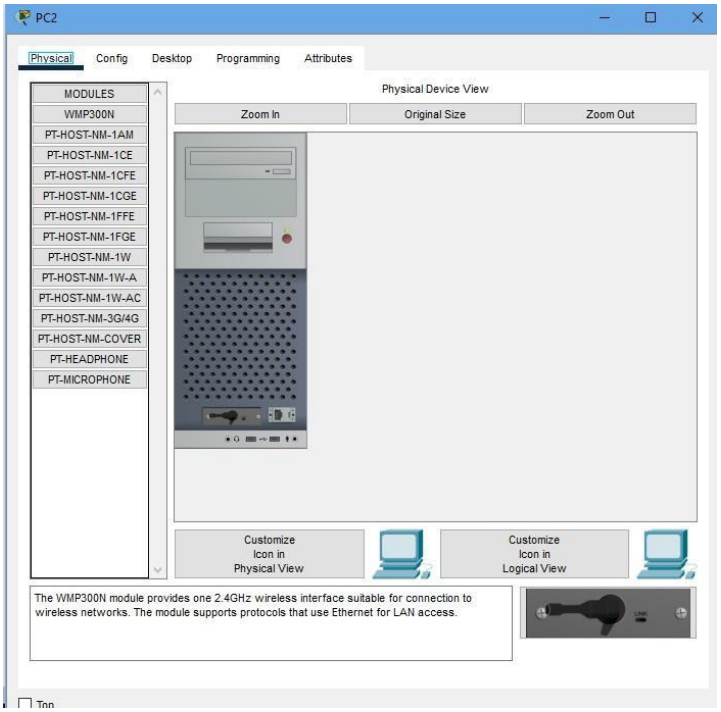
4. Configuration of Wireless Network



In Packet Tracer, PCs do not have a wireless port by default. We have to add it physically. We will add the WMP300N module on the PC.

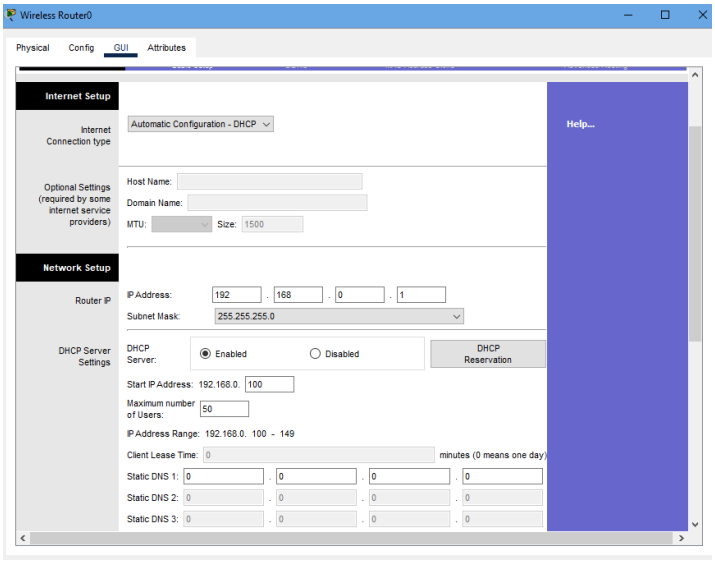
Do the following steps to add WMP300N module:

1. Go to the Physical tab.
2. Turn off the PC.
3. Zoom out.
4. Remove the existing module from the bottom.
5. Add the WMP300N module in its place.
6. Turn on the PC.



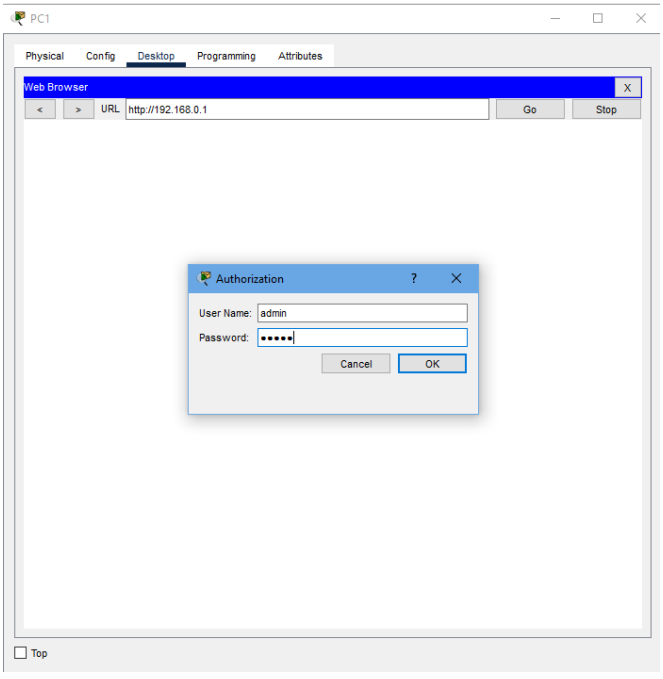
Go to GUI of wireless router.

1. DHCP is configured and enabled on Wireless router.
2. IP pool for DHCP is 192.168.0.100 to 192.168.0.150.
3. PCs are configured to receive IP from DHCP Server.
4. No security is configured.
5. Default SSID is configured to Default.
6. Topology is working on infrastructure mode.
7. Default user name and password (on router) is admin.
8. IP of wireless router is set to 192.168.0.1.

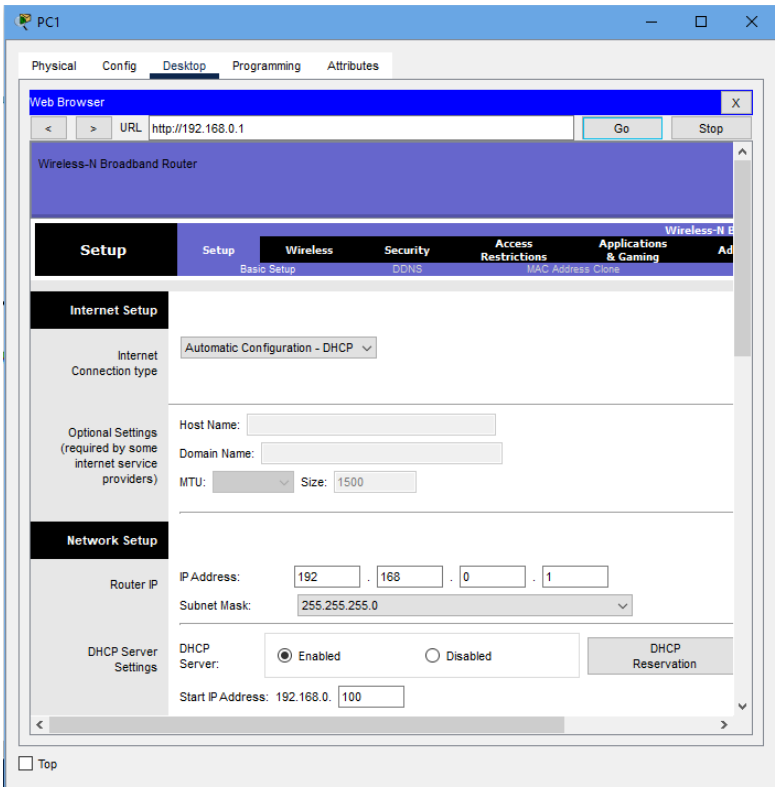


Now, enable DHCP on PCs.

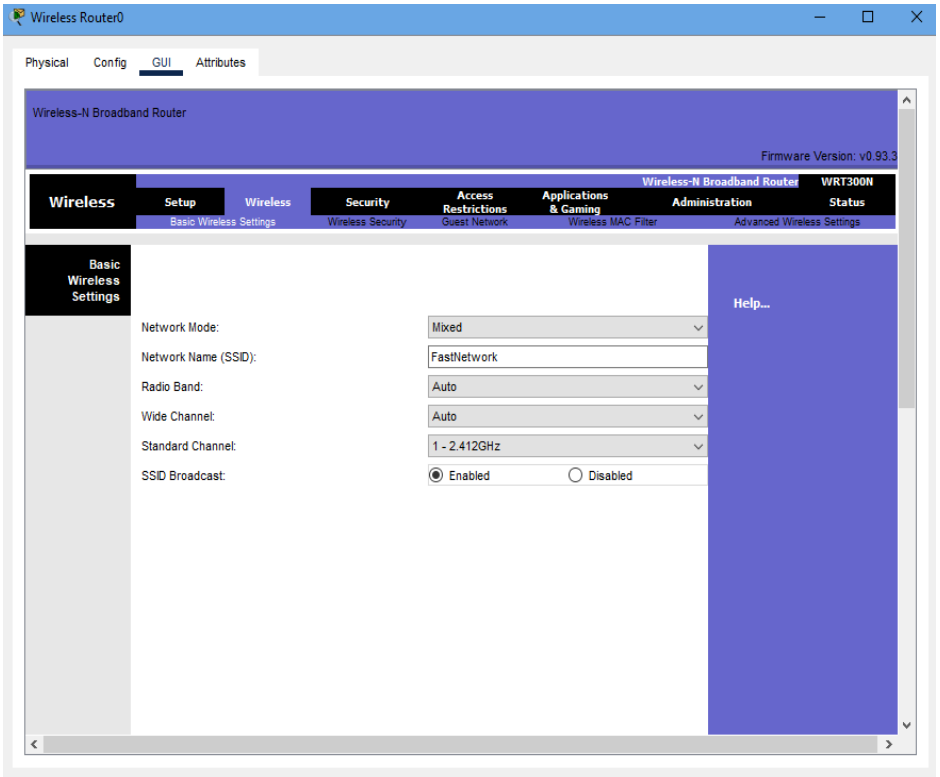
Click on any PC and select Web Browser. As given, IP of Wireless router is 192.168.0.1 so give it in Web browser and press enter, now it will ask for authentication which is also given above. Default username is admin and password is also admin.



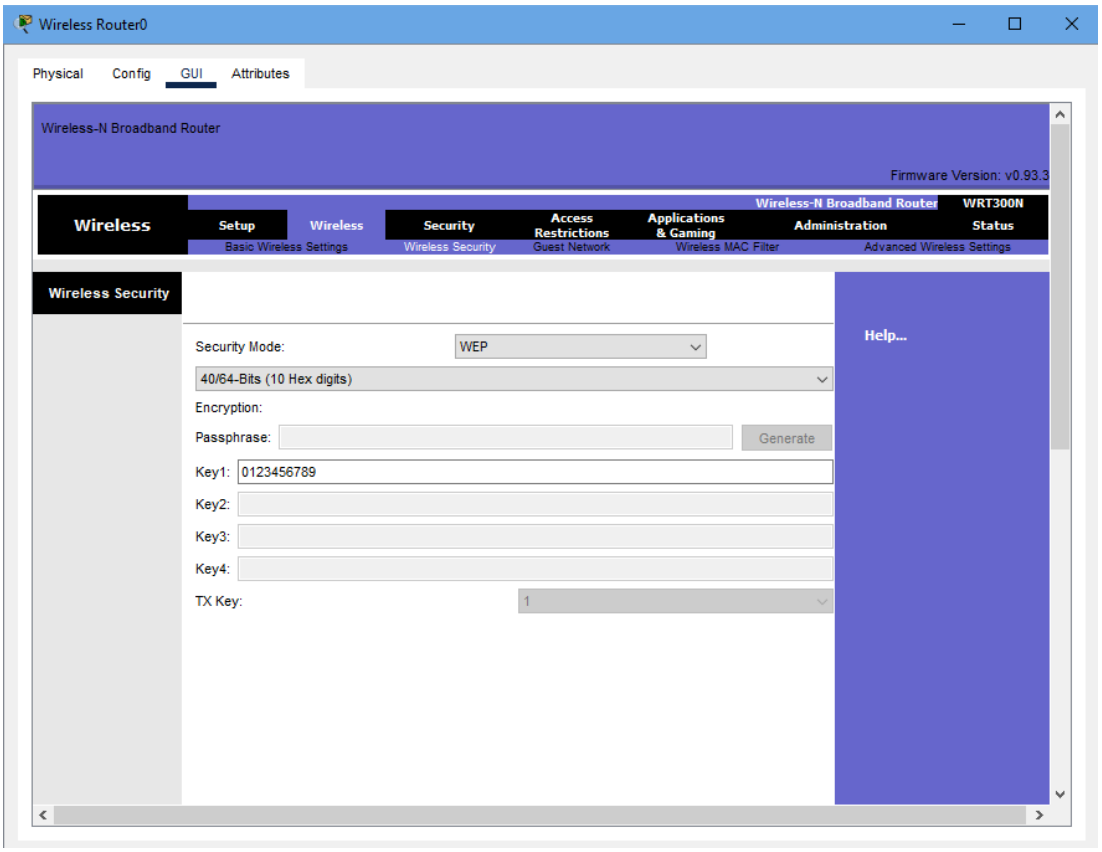
This will bring GUI mode of Wireless router.



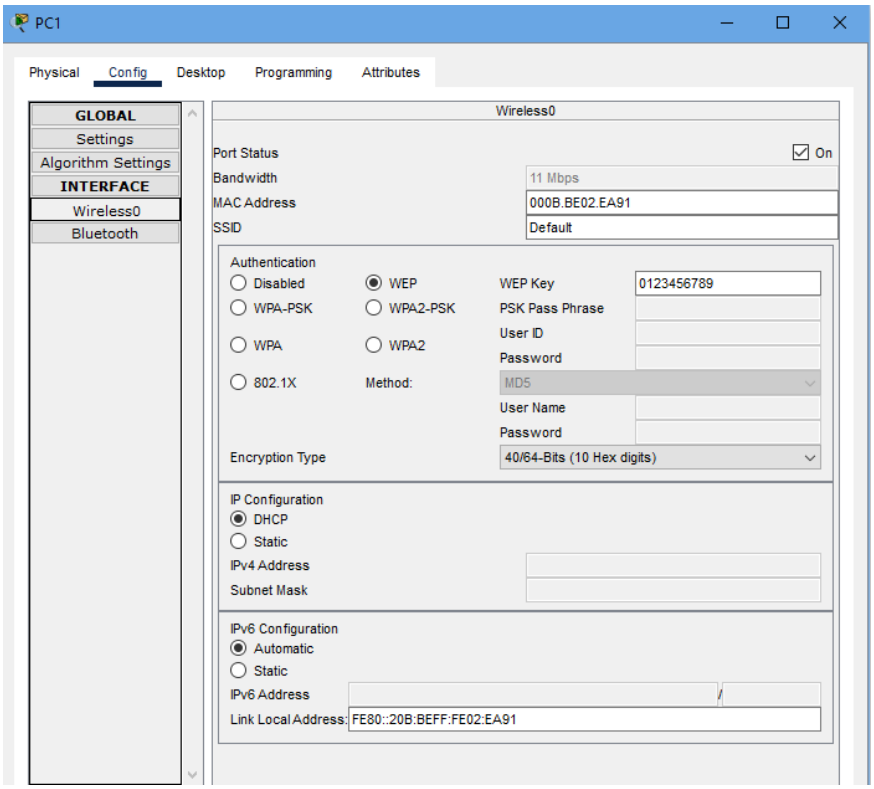
Open wireless router, now click on wireless tab in the router and set default SSID to **FastNetwork** (you can name it something else too). Go at the end of page and click on Save Setting.



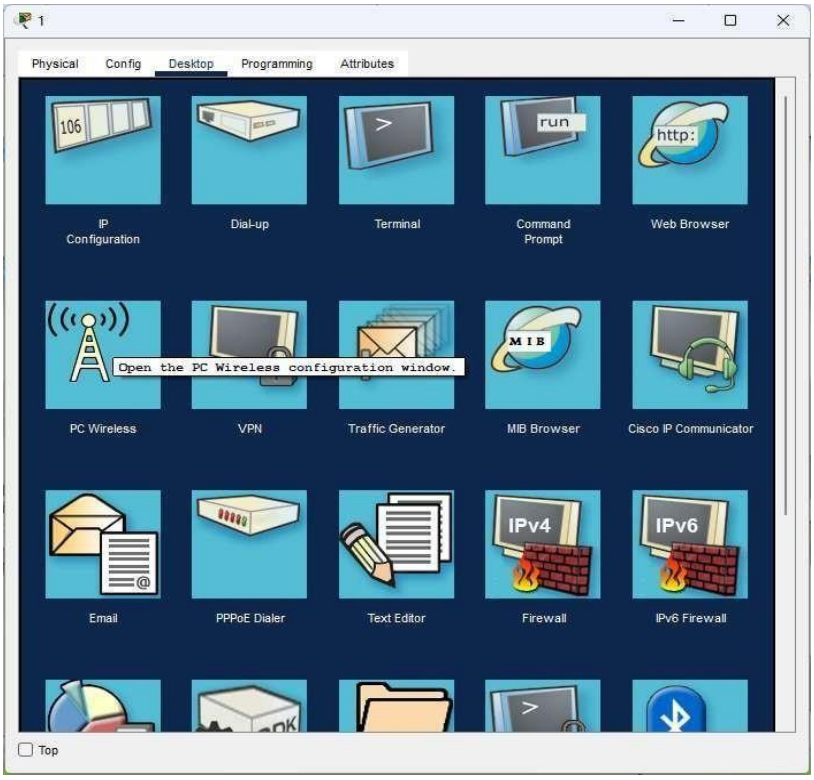
Now Select wireless security and change Security Mode to WEP. Generate key1 and click on save setting.



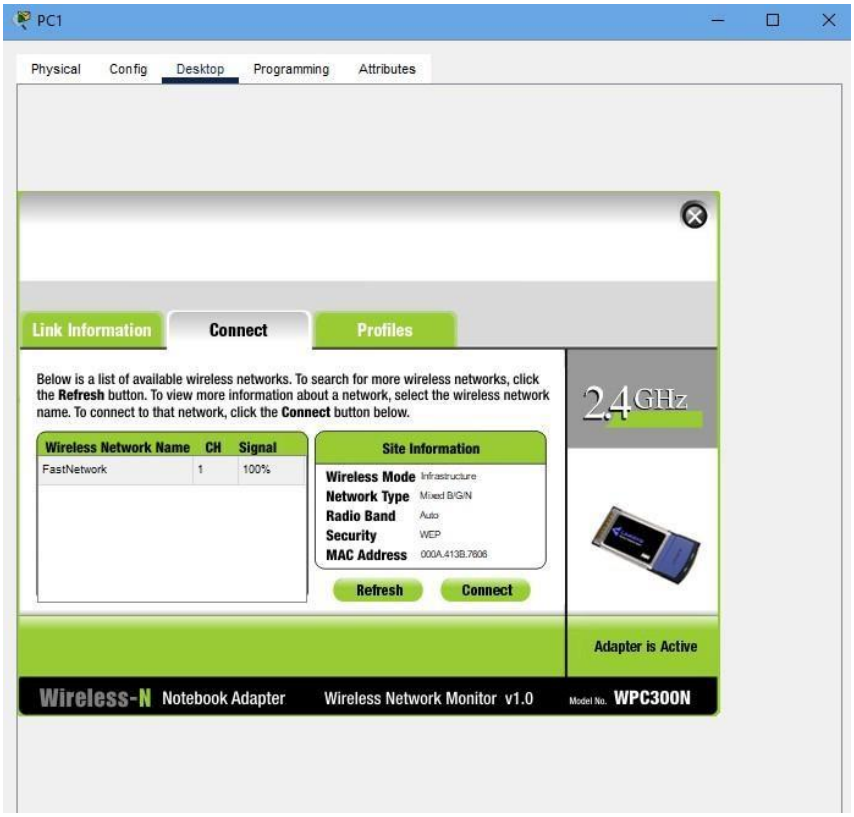
Go to PC interface wireless0, Specify WEP Key on every PC.



Now it's time to connect PCs from Wireless router. Click PC, go to PC Wireless:



Click on connect tab. It will ask for WEP key, insert 0123456789 and click connect.

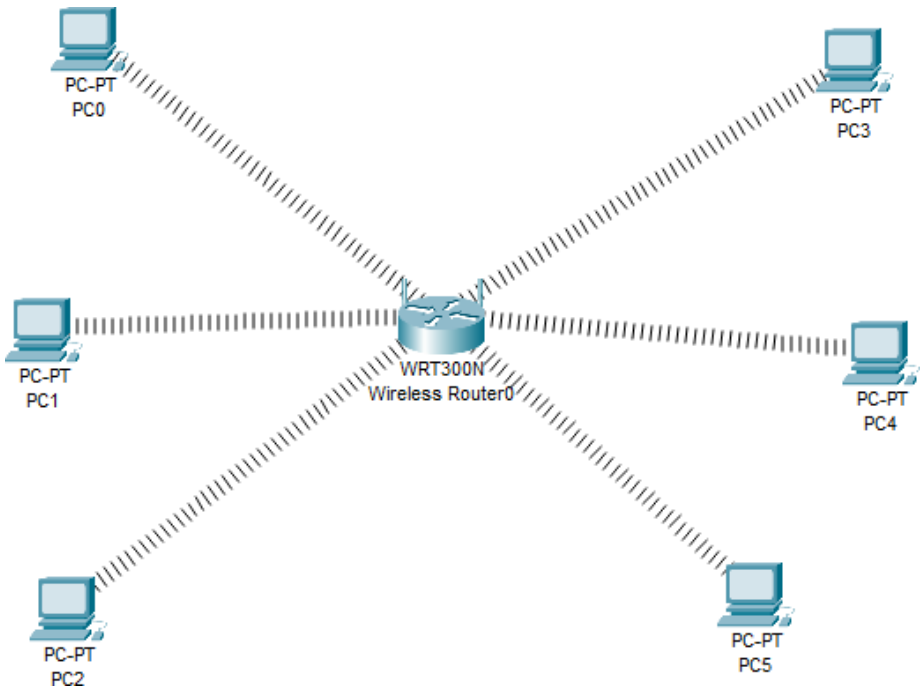


It will connect you with wireless router as shown below:



Repeat this process on all PCs.

Now your network will become a wireless network as shown below.



NAT

1. Introduction to NAT and its type

Network Address Translation (NAT) is a mechanism by which a device modifies a packet's TCP/IP address/port number and maps the IP address from one realm to another (usually from private IP address to public IP address and vice versa). This is accomplished by the NAT device allocating a temporary port number on the public side of the NAT when forwarding outbound packets from the internal host to the Internet, maintaining this mapping for a predetermined period of time, and forwarding inbound packets received from the Internet on this public port back to the internal host.

NAT devices are used to avoid the exhaustion of IPv4 address space by allowing multiple hosts to share a single public/Internet address. Also due to its mapping nature (i.e., a mapping can only be created by a transmission from an internal host), NAT device is preferred to be installed even when IPv4 address exhaustion is not a problem (for example when there is only one host at home), to provide some sort of security/shield for the internal hosts against threats from the Internet.

Despite the fact that NAT provides some shields for the internal network, one must distinguish NAT solution from firewall solution. NAT is not a firewall solution. A firewall is a security solution designed to enforce the security policy of an organization, while NAT is a connectivity solution to allow multiple hosts to use a single public IP address. Understandably both functionalities are difficult to separate at times, since many (typically consumer) products claims to do both with the same device and simply label the device a “NAT box”. But we do want to make this distinction rather clear, as PJNATH is a NAT traversal helper and not a firewall bypass solution (yet).

Following are the types of NAT:

1. **Static NAT (Network Address Translation):** Static NAT (Network Address Translation) is one-to-one mapping of a private IP address to a public IP address. Static NAT (Network Address Translation) is useful when a network device inside a private network needs to be accessible from internet.

2. **Dynamic NAT (Network Address Translation):** Dynamic NAT can be defined as mapping of a private IP address to a public IP address from a group of public IP addresses called as NAT pool. Dynamic NAT establishes a one-to-one mapping between a private IP address to a public IP address. Here the public IP address is taken from the pool of IP addresses configured on the end NAT router. The public to private mapping may vary based on the available public IP address in NAT pool.

3. **PAT (Port Address Translation):** Port Address Translation (PAT) is another type of dynamic NAT which can map multiple private IP addresses to a single public IP address by using a technology known as Port Address Translation.

2. Configuration of NAT

Static NAT Configuration

Static NAT is used to do a one-to-one mapping between an inside address and an outside address. Static NAT also allows connections from an outside host to an inside host. Usually, static NAT is used for servers inside your network. For example, you may have a web server with the inside IP address 192.168.0.10 and you want it to be accessible when a remote host makes a request to 209.165.200.10. For this to work, you must do a static NAT mapping between those to IPs. In this example, we will use the Fast Ethernet 0/1 as the inside NAT interface, the interface connecting to our network, and the Serial 0/0/0 interface as the outside NAT interface, the one connecting to our service provider.

Working Steps on Router:

Router(config)#ip nat inside source static 192.168.0.10 209.165.200.10
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside
Router(config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

Dynamic NAT Configuration

Dynamic NAT is used when you have a “pool” of public IP addresses that you want to assign to your internal hosts dynamically. Don’t use dynamic NAT for servers or other devices that need to be accessible from the Internet. In this example, we will define our internal network as 192.168.0.0/24. We also have the pool of public IP addresses from 209.165.200.226 to 209.165.200.240 and our assigned netmask is 255.255.255.224. When you configure dynamic NAT, you have to define an ACL to permit only those addresses that are allowed to be translated.

Working Steps on Router:

Router(config)#ip nat pool NAT-POOL 209.165.200.226 209.165.200.240 netmask 255.255.255.224
Router(config)#access-list 1 permit 192.168.0.0 0.255.255.255
Router(config)#ip nat inside source list 1 pool NAT-POOL
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside
Router (config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

3. NAT Overload (PAT) Configuration

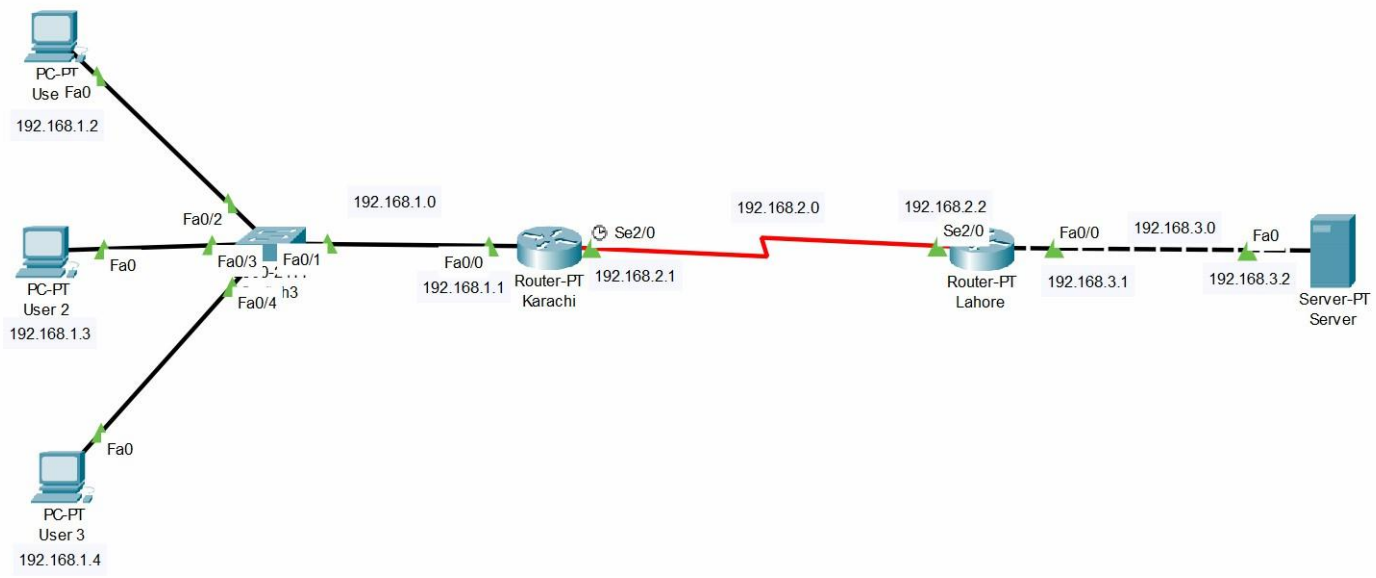
NAT Overload, sometimes also called PAT, is probably the most used type of NAT. You can configure NAT overload in two ways, depending on how many public IP address you have available. The first case, and one of the most often seen cases, is that you have only one public IP address allocated by your ISP. In this case, you map all your inside hosts to the available IP address. The configuration is almost the same as for dynamic NAT, but this time you specify the outside interface instead of a NAT pool.

Working Steps on Router:

Router(config)#access list 1 permit 192.168.0.0 0.255.255.255
Router(config)#ip nat inside source list 1 interface serial 0/0/0 overload
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside
Router(config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

4. Implementation

In this example we configure NAT overload (PAT) on the router which is located in Karachi and static NAT is configured on Lahore router. A simple topology is shown below:



Assigning IP address to Karachi Router:

```
Router>enable
Router# configure terminal
Router(config)#hostname R1
 R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface Serial2/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#clock rate 64000
R1(config-if)#bandwidth 64
 R1(config-if)#no shutdown
R1(config-if)#exit
```

Assigning IP address to Lahore Router:

```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface Serial2/0
R2(config-if)#ip address 192.168.2.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
```

Configuring NAT Overload (PAT) on router Karachi:

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.2 0 0.0.0.0
R1(config)#access-list 1 permit 192.168.1.3 0.0.0.0
R1(config)#access-list 1 deny any
R1(config)#ip nat pool ccna 50.0.0.1 50.0.0.1 netmask 255.0.0.0
 R1(config)#ip nat inside source list 1 pool ccna overload
R1(config)#interface FastEthernet 0/0
```

R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#interface Serial 0/0/0
R1(config-if)#ip nat outside
R1(config-if)#exit

Configuring Static NAT on router Lahore:

R2>enable
R2#configure terminal
R2(config)#ip nat inside source static 192.168.3.1 200.0.0.10
R2(config)#interface Serial 2/0
R2(config-if)#ip nat outside
R2(config-if)#exit
R2(config)#interface FastEthernet 0/0
R2(config-if)#ip nat inside
R2(config-if)#exit

Configuring static route on routers:
R1(config)#ip route 200.0.0.0 255.255.255.0 100.0.0.2
R2(config)#ip route 50.0.0.0 255.0.0.0 100.0.0.1

Now ping server from a PC or open website through web browser using 200.0.0.10 IP address. To check the connectivity after configuring NAT. Use show “ip nat translation command” to verify NAT implementation on router.

```
R1#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 50.0.0.1:1        10.0.0.20:1        200.0.0.10:1       200.0.0.10:1
icmp 50.0.0.1:2        10.0.0.20:2        200.0.0.10:2       200.0.0.10:2
icmp 50.0.0.1:3        10.0.0.20:3        200.0.0.10:3       200.0.0.10:3
icmp 50.0.0.1:4        10.0.0.20:4        200.0.0.10:4       200.0.0.10:4
tcp 50.0.0.1:1024      10.0.0.10:1025     200.0.0.10:80      200.0.0.10:80
tcp 50.0.0.1:1025      10.0.0.20:1025     200.0.0.10:80      200.0.0.10:80
```

5. Wireshark of NAT

In this section, we’ll capture packets from a simple web request from a client PC in a home network to a www.google.com server.

Within the home network, the home network router provides a NAT service. Above topology shows our Wireshark trace-collection scenario. As in our other Wireshark labs, we collect a Wireshark trace on the client PC in our home network. This file is called NAT_home_side. Because we are also interested in the packets being sent by the NAT router into the ISP, we’ll collect a second trace file at a PC (not shown) tapping into the link from the home router into the ISP network, as shown in topology. (The hub device shown on the ISP side of the router is used to tap into the link between the NAT router and the first hop router in the ISP). Client-to-server packets captured by Wireshark at this point will have undergone NAT translation. The Wireshark trace file captured on the ISP side of the home router is called

NAT_ISP_side. Open the NAT_home_side file and answer the following questions. You might find it useful to use a Wireshark filter so that only frames containing HTTP messages are displayed from the trace file. Whenever possible, when answering a question below, you should hand in a printout of the packet(s) within the trace that you used to answer the question asked. Annotate the printout to explain your answer. To print a packet, use File->Print, choose Selected packet only, choose Packet summary line, and select the minimum amount of packet detail that you need to answer the given in the Exercise Section of NAT.

Download the zip file http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip and extract the files need for this lab.